Reinforcement Learning from Diverse Human Preferences

Summary

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1 Problem Setup

Traditional reinforcement learning from human preferences assumes a single, consistent reward function that represents human preferences. However, in reality, different humans have diverse preferences that may conflict or change over time. This paper addresses the challenge of learning from diverse human preferences, where multiple people with different values and priorities provide feedback to the same RL agent.

2 Thesis

The central thesis is that reinforcement learning agents can effectively learn from diverse human preferences by using latent space constraints and confidence-based model ensembling. The authors argue that by controlling the divergence of reward representations and using confidence-weighted aggregation, agents can handle conflicting feedback while maintaining stable learning.

3 Key Concepts

Diverse Preferences: Human preferences that vary across individuals, potentially containing conflicts or contradictions that must be reconciled by the learning system.

Latent Space Constraints: Restrictions placed on the reward representation in latent space to prevent divergence and maintain stability when learning from conflicting feedback.

Model Ensembling: The process of combining multiple reward models, with confidence-based weighting to give more influence to more reliable models.

Preference-Based RL: Learning reward functions from human preferences rather than explicit reward signals, typically through pairwise comparisons.

Confidence-Based Aggregation: A method for combining diverse reward models where each model's influence is weighted by its confidence or reliability.

4 Methods

The authors develop a method for learning from diverse human preferences:

1. **Latent Space Regularization:** They impose strong constraints on the latent space of reward representations to control divergence and maintain stability.

2. **Confidence-Based Ensembling:** Multiple reward models are trained from different human feedback sources and combined using confidence-weighted averaging.

3. **Stability Mechanisms:** The system includes mechanisms to handle conflicting feedback without destabilizing the learning process.

4. **Preference Collection:** Human preferences are collected through standard pairwise comparison interfaces but from multiple diverse users.

5. **Policy Optimization:** The learned reward ensemble is used to train policies using standard RL algorithms like PPO.

The key innovation is the combination of latent space constraints with confidence-based aggregation to handle preference diversity.

5 Results

The authors demonstrate their approach on complex robotics and locomotion tasks:

- **Stability:** The method maintains stable learning even when receiving conflicting feedback from different humans.
- **Performance:** Significantly outperforms existing preference-based RL algorithms when learning from diverse feedback sources.
- **Robustness:** The system is robust to variations in human feedback quality and consistency across different users.
- **Complex Tasks:** Successfully learns complex behaviors in locomotion and manipulation tasks despite diverse and sometimes contradictory human preferences.
- **Scalability:** The approach scales to multiple human feedback sources without requiring significant modifications to the underlying RL algorithms.
- **Generalization:** Policies learned from diverse preferences generalize well to new situations and maintain performance across different evaluation scenarios.

The results show that the method can effectively reconcile diverse human preferences while maintaining stable and efficient learning.

6 Significance

This work addresses an important challenge in human-AI collaboration:

- **Real-World Applicability:** Makes preference-based RL more practical by handling the reality of diverse human preferences rather than assuming consensus.
- **Stability Advances:** Provides mechanisms for maintaining learning stability when faced with conflicting or inconsistent human feedback.
- **Scalable Collaboration:** Enables multiple humans to collaborate on training AI systems without requiring them to have identical preferences.
- **Robustness:** Improves the robustness of preference-based learning systems to variations in human feedback quality and consistency.

- **Broader Impact:** Facilitates the deployment of RL systems in real-world scenarios where multiple stakeholders with different values need to be considered.

The paper represents an important step toward making preference-based reinforcement learning more practical and robust for real-world applications where human diversity is the norm rather than the exception.